

Annuity practice speedwork

Rules: You will have *exactly 5 minutes* to answer each question. After the time is up, you will work with a partner to compare solutions.

Problem.

You are given a perpetual annuity-immediate with annual payments increasing in geometric progression, with a common ratio of 1.07. The annual effective interest rate is 12%. The first payment is 1.

Calculate the present value of this annuity.

- A. 18
- B. 19
- C. 20
- D. 21
- E. 22

Problem.

A bank lends \$100,000 to Sam. The loan is repaid with level payments at the end of each year for 30 years based on an annual effective interest rate of 5%.

The bank reinvests the loan payments at an annual effective interest rate of 4%.

Calculate the bank's annual effective yield rate over the 30-year period.

- A. 4.0%
- B. 4.1%
- C. 4.2%
- D. 4.4%
- E. 4.7%

Problem.

John receives a perpetuity paying 2 at the end of the 4th year, 4 at the end of the 6th year, 6 at the end of the 8th year, etc.

The present value of this perpetuity at an annual effective interest rate of 10% equals X .

Calculate X .

- A. 35
- B. 37
- C. 40
- D. 42
- E. 45

Problem.

Steve purchases an annuity-immediate that pays \$1,500 at the end of each year for 15 years.

Jonathan purchases an annuity-immediate that pays X at the end of each month for 15 years.

Nancy purchases a perpetuity that makes continuous payments at a rate of X per year.

At an annual effective interest rate of 6%, the present value of Steve's annuity is equal to the present value of Jonathan's annuity.

Calculate the price that Nancy pays for the perpetuity, assuming the same interest rate.

- A. 1,984
- B. 2,028
- C. 2,088
- D. 2,304
- E. 2,496

Problem.

Gloria borrows \$100,000 to be repaid over 30 years. You are given:

- i. Her first payment is X at the end of year 1.
- ii. Her payments increase at the rate of \$100 per year for the next 19 years and remain level for the following 10 years.
- iii. The effective rate of interest is 5% per annum.

Calculate X .

- A. 5505
- B. 5555
- C. 5605
- D. 5655
- E. 5665